

Phys 115B HW 6

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1 D

Question 1. 1 Normalization (Griffiths 5.4, same)

(a) If ψ_a and ψ_b are orthogonal, and both are normalized, what is the constant A in equation $\psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) = A(\psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2) \pm \psi_b(\mathbf{r}_1)\psi_a(\mathbf{r}_2))$?

(b) If $\psi_a = \psi_b$ (and it is normalized), what is A ? (This case, of course, occurs only for bosons.)

Proof.

(a) Consider the normalization condition of individual states, we get the following equation:

$$\forall u, v \in \{a, b\}, \quad \langle \psi_u | \psi_v \rangle = \delta_{uv} \quad (1)$$

Which, if consider the states as tensor product, then one has the following (we'll use $|\psi\rangle_1$ to indicate $\mathbf{r}_1$, and $|\psi\rangle_2$ to indicate $\mathbf{r}_2$):

$$|\psi_{\pm}\rangle = A(|\psi_a\rangle_1 \otimes |\psi_b\rangle_2 \pm |\psi_b\rangle_1 \otimes |\psi_a\rangle_2) \quad (2)$$

So, one gets the following regarding the normalization condition:

$$1 = \langle \psi_{\pm} | \psi_{\pm} \rangle \quad (3)$$

$$= |A|^2 (\langle \psi_a | \psi_a \rangle_1 \cdot \langle \psi_b | \psi_b \rangle_2 + \langle \psi_b | \psi_b \rangle_1 \cdot \langle \psi_a | \psi_a \rangle_2 \pm \langle \psi_b | \psi_a \rangle_1 \cdot \langle \psi_a | \psi_b \rangle_2 \pm \langle \psi_a | \psi_b \rangle_1 \cdot \langle \psi_b | \psi_a \rangle_2) \quad (4)$$

$$= |A|^2 (1 \cdot 1 + 1 \cdot 1 \pm 0 \cdot 0 \pm 0 \cdot 0) = 2|A|^2 \quad (5)$$

So, one gets that $|A| = \frac{1}{\sqrt{2}}$. Take the positive real value, take $A = \frac{1}{\sqrt{2}}$ is the normalization constant.

(b) If $\psi = \psi_a = \psi_b$, then the state $\psi_+ = 2A\psi(\mathbf{r}_1)\psi(\mathbf{r}_2)$. Which, consider the normalization, one has the following:

$$1 = \langle \psi_+ | \psi_+ \rangle = 4|A|^2 \langle \psi | \psi \rangle_1 \cdot \langle \psi | \psi \rangle_2 = 4|A|^2 \quad (6)$$

So, one gets $|A|^2 = \frac{1}{4}$, or $|A| = \frac{1}{2}$. Take A as positive real, one has $A = \frac{1}{2}$.

□

2 D

Question 2. 2 Particles in the infinite square well (Griffiths 5.5, same)

- (a) Write down the Hamiltonian for two noninteracting identical particles in the infinite square well. Verify that the fermion ground state given by $\frac{\sqrt{2}}{a} \left(\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right)$ is an eigenfunction of $\hat{H}$, with the appropriate eigenvalue.
- (b) Find the next two excited states (beyond the ones given in the example)–wave functions, energies, and degeneracies–for each of the three cases (distinguishable, identical bosons, identical fermions).

Hamiltonian (when restricting $x_1, x_2 \in [0, a]$):

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} \right)$$

Proof.

- (a) Applying $\hat{H}$ given above to the given fermion ground state, we get:

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} \right) \left(\frac{\sqrt{2}}{a} \left(\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \right) \quad (7)$$

$$= -\frac{\hbar^2}{\sqrt{2}am} \left(-\frac{\pi^2}{a^2} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) + \frac{4\pi^2}{a^2} \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \quad (8)$$

$$- \frac{\hbar^2}{\sqrt{2}am} \left(-\frac{4\pi^2}{a^2} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) + \frac{\pi^2}{a^2} \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \quad (9)$$

$$= \frac{5\hbar^2\pi^2}{\sqrt{2}a^3m} \left(\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \quad (10)$$

$$= \frac{5\hbar^2\pi^2}{2ma^2} \cdot \left(\frac{\sqrt{2}}{a} \left(\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \right) \quad (11)$$

This verifies that the given function is indeed an eigenfunction of $\hat{H}$, with eigenvalue $\frac{5\hbar^2\pi^2}{2ma^2}$.

- (b) Here are the cases for each case:

– **Distinguishable:**

Since the distinguishable states are $\psi_{n_1 n_2} = \psi_{n_1}(x_1)\psi_{n_2}(x_2)$ with eigenvalue $E_{n_1 n_2} = (n_1^2 + n_2^2)K$ (for $K = \frac{\hbar^2\pi^2}{2ma^2}$). With the case of $(n_1, n_2) = (1, 1), (1, 2), (2, 1)$ being described (corresponds to $n_1^2 + n_2^2 = 2, 5, 5$ respectively), the next two eigenvalues should be related to $(n_1, n_2) = (2, 2)$ (corresponding to $E_{22} = 8K$), and $(n_1, n_2) = (1, 3), (3, 1)$ (corresponding to $E_{13} = E_{31} = 10K$). Which, the informations are as below:

$$\text{Wave function: } \frac{2}{a} \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right), \quad \text{Eigenvalue: } E_{22} = 8K, \quad \text{Degeneracy: } 1 \quad (12)$$

$$\text{Wave function: } \frac{2}{a} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{3\pi x_2}{a}\right) \text{ or } \frac{2}{a} \sin\left(\frac{3\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \quad (13)$$

$$\text{Eigenvalue: } E_{13} = E_{31} = 10K, \quad \text{Degeneracy: } 2 \quad (14)$$

– **Identical bosons:**

Since the identical bosons are given by the following, and with energy $E_{n_1 n_2} = (n_1^2 + n_2^2)K$ again:

$$\psi_{n_1 n_2} = \begin{cases} \frac{\sqrt{2}}{a} (\psi_{n_1}(x_1)\psi_{n_2}(x_2) + \psi_{n_1}(x_2)\psi_{n_2}(x_1)) & n_1 \neq n_2 \\ \frac{2}{a} \psi_{n_1}(x_1)\psi_{n_1}(x_2) & n_1 = n_2 \end{cases} \quad (15)$$

With the case of $(n_1, n_2) = (1, 1), (1, 2), (2, 1)$ being mentioned (corresponds to $n_1^2 + n_2^2 = 2, 5, 5$, and the case $(1, 2), (2, 1)$ has identical wavefunctions), then the next two eigenvalues should be related to $(n_1, n_2) = (2, 2)$ (corresponds to $E_{22} = 8K$), and $(n_1, n_2) = (1, 3), (3, 1)$ (correspond to $E_{13} = E_{31} = 10K$). As a side note, $(1, 3), (3, 1)$ by symmetric tensor are the same. So, the informations are as below:

$$\text{Wave function: } \frac{2}{a} \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right), \quad \text{Eigenvalue: } E_{22} = 8K, \quad \text{Degeneracy: } 1 \quad (16)$$

$$\text{Wave function: } \frac{\sqrt{2}}{a} \left(\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{3\pi x_2}{a}\right) + \sin\left(\frac{3\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \quad (17)$$

$$\text{Eigenvalue: } E_{13} = 10K, \quad \text{Degeneracy: } 1 \quad (18)$$

– **Identical fermions:**

Since the identical fermions are given by the following, and with energy $E_{n_1 n_2} = (n_1^2 + n_2^2)K$ again:

$$\psi_{n_1 n_2} = \frac{\sqrt{2}}{a} (\psi_{n_1}(x_1)\psi_{n_2}(x_2) - \psi_{n_1}(x_2)\psi_{n_2}(x_1)) \quad (19)$$

Hence, the allowed nonzero states only happen with $n_1 \neq n_2$.

With the case of $(n_1, n_2) = (1, 2), (2, 1)$ being described (corresponds to $n_1^2 + n_2^2 = 5, 5$ respectively, and in this case the two wave functions differ by a factor of -1 , which is a global phase instead; below we'll just choose (n_1, n_2) once, and not consider (n_2, n_1)), then next two eigenvalues should be related to $(n_1, n_2) = (1, 3)$ (corresponding to $E_{13} = 10K$) and $(n_1, n_2) = (2, 3)$ (corresponding to $E_{23} = 13K$). Which, the informations are as below:

$$\text{Wave function: } \frac{\sqrt{2}}{a} \left(\sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{3\pi x_2}{a}\right) - \sin\left(\frac{3\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \right) \quad (20)$$

$$\text{Eigenvalue: } E_{13} = 10K, \quad \text{Degeneracy: } 1 \quad (21)$$

$$\text{Wave function: } \frac{\sqrt{2}}{a} \left(\sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{3\pi x_2}{a}\right) - \sin\left(\frac{3\pi x_1}{a}\right) \sin\left(\frac{2\pi x_2}{a}\right) \right) \quad (22)$$

$$\text{Eigenvalue: } E_{23} = 13K, \quad \text{Degeneracy: } 1 \quad (23)$$

□

3 D

Question 3. 3 Exchange forces across orbitals (Griffiths 5.6, same)

Imagine two noninteracting particles, each of mass m , in the infinite square well. If one is in the state ψ_n , and the other in state ψ_l ($l \neq n$), calculate $\langle (x_1 - x_2)^2 \rangle$, assuming:

- (a) They are distinguishable particles
- (b) They are identical bosons
- (c) They are identical fermions

Proof.

- (a) As distinguishable particles, the wave functions are $\psi_{nl} = \frac{2}{a} \sin\left(\frac{n\pi x_1}{a}\right) \sin\left(\frac{l\pi x_2}{a}\right)$. Which, one has the following:

$$\langle (x_1 - x_2)^2 \rangle = \int_{x_1=0}^a \int_{x_2=0}^a \psi_{nl}^* (x_1 - x_2)^2 \psi_{nl} dx_1 dx_2 \quad (24)$$

$$= \frac{4}{a^2} \int_{x_1=0}^a x_1^2 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (25)$$

$$+ \frac{4}{a^2} \int_{x_1=0}^a \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2^2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (26)$$

$$- \frac{8}{a^2} \int_{x_1=0}^a x_1 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (27)$$

$$= \frac{2}{a} \int_0^a x_1^2 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 + \frac{2}{a} \int_0^a x_2^2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 - \frac{8}{a^2} \cdot \left(\frac{a^2}{4}\right)^2 \quad (28)$$

$$= \left(\frac{a^2}{3} - \frac{a^2}{2n^2\pi^2}\right) + \left(\frac{a^2}{3} - \frac{a^2}{2l^2\pi^2}\right) - \frac{a^2}{2} \quad (29)$$

$$= \left(\frac{1}{6} - \frac{1}{2n^2\pi^2} - \frac{1}{2l^2\pi^2}\right) a^2 \quad (30)$$

$$= \frac{n^2 l^2 \pi^2 - 3l^2 \pi^2 - 3n^2 \pi^2}{6n^2 l^2 \pi^2} a^2 \quad (31)$$

(Rest on the next few pages)

(b),(c) As identical bosons/fermions (with $l \neq n$), the wavefunction is $\psi_{nl} = \frac{\sqrt{2}}{a} \left(\sin\left(\frac{n\pi x_1}{a}\right) \sin\left(\frac{l\pi x_2}{a}\right) \pm \sin\left(\frac{n\pi x_2}{a}\right) \sin\left(\frac{l\pi x_1}{a}\right) \right)$ (+ for bosons, and - for fermions)

Which, one has the following:

$$\langle (x_1 - x_2)^2 \rangle = \int_{x_1=0}^a \int_{x_2=0}^a \psi_{nl}^*(x_1 - x_2)^2 \psi_{nl} dx_1 dx_2 \quad (32)$$

$$(33)$$

$$= \frac{2}{a^2} \int_{x_1=0}^a x_1^2 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (34)$$

$$\pm \frac{4}{a^2} \int_{x_1=0}^a x_1^2 \sin\left(\frac{n\pi x_1}{a}\right) \sin\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a \sin\left(\frac{n\pi x_2}{a}\right) \sin\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (35)$$

$$+ \frac{2}{a^2} \int_{x_1=0}^a \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2^2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (36)$$

$$\pm \frac{4}{a^2} \int_{x_1=0}^a \sin\left(\frac{n\pi x_1}{a}\right) \sin\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2^2 \sin\left(\frac{n\pi x_2}{a}\right) \sin\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (37)$$

$$- \frac{4}{a^2} \int_{x_1=0}^a x_1 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (38)$$

$$\mp \frac{8}{a^2} \int_{x_1=0}^a x_1 \sin\left(\frac{n\pi x_1}{a}\right) \sin\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin\left(\frac{n\pi x_2}{a}\right) \sin\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (39)$$

$$+ \frac{2}{a^2} \int_{x_1=0}^a x_1^2 \sin^2\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a \sin^2\left(\frac{n\pi x_2}{a}\right) dx_2 \quad (40)$$

$$+ \frac{2}{a^2} \int_{x_1=0}^a \sin^2\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2^2 \sin^2\left(\frac{n\pi x_2}{a}\right) dx_2 \quad (41)$$

$$- \frac{4}{a^2} \int_{x_1=0}^a x_1 \sin^2\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin^2\left(\frac{n\pi x_2}{a}\right) dx_2 \quad (42)$$

$$(43)$$

$$= \frac{2}{a} \int_{x_1=0}^a x_1^2 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 + \frac{2}{a} \int_{x_2=0}^a x_2^2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (44)$$

$$- \frac{4}{a^2} \int_{x_1=0}^a x_1 \sin^2\left(\frac{n\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin^2\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (45)$$

$$\mp \frac{8}{a^2} \int_{x_1=0}^a x_1 \sin\left(\frac{n\pi x_1}{a}\right) \sin\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin\left(\frac{n\pi x_2}{a}\right) \sin\left(\frac{l\pi x_2}{a}\right) dx_2 \quad (46)$$

$$- \frac{4}{a^2} \int_{x_1=0}^a x_1 \sin^2\left(\frac{l\pi x_1}{a}\right) dx_1 \int_{x_2=0}^a x_2 \sin^2\left(\frac{n\pi x_2}{a}\right) dx_2 \quad (47)$$

$$(48)$$

$$= a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right) + a^2 \left(\frac{1}{3} - \frac{1}{2l^2\pi^2} \right) - \frac{a^2}{2} \mp \frac{32n^2l^2((-1)^{n-l} - 1)^2 a^2}{\pi^4(n^2 - l^2)^4} \quad (49)$$

$$= a^2 \left(\frac{1}{6} - \frac{1}{2n^2\pi^2} - \frac{1}{2l^2\pi^2} \mp \frac{32n^2l^2((-1)^{n-l} - 1)^2}{\pi^4(n^2 - l^2)^4} \right) \quad (50)$$

Which, - corresponds to identical bosons, while + corresponds to identical fermions.

□

Question 4. 4 Three particles (Griffiths 5.8, 5.7 in second edition)

Suppose you had three particles, one in state $\psi_a(x)$, one in state $\psi_b(x)$, and one in state $\psi_c(x)$. Assuming ψ_a, ψ_b , and ψ_c are orthonormal, construct the three-particle states representing:

- (a) Distinguishable particles
- (b) Identical bosons
- (c) Identical fermions.

Keep in mind that (b) must be completely symmetric, under interchange of any pair of particles, and (c) must be completely anti-symmetric, in the same sense.

Proof. Here we'll assume the grader has some background in group theory (because this simplifies the terms a lot).

- (a) For three distinguishable particles, one has the wave function $\psi_{abc} = \psi_a(x_1)\psi_b(x_2)\psi_c(x_3)$.
- (b) For identical bosons, one has the wave function being some multiples of $\sum_{\sigma \in S_3} \psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})$ (where S_3 denotes the permutation group of $\{1, 2, 3\}$, there's too many terms I'll abbreviate it as this). The reason is simply because such sum is symmetric under any possible permutation of $\{1, 2, 3\}$ (since the left multiplication of group element is a bijection on the group itself).
Which, notice that if $\sigma, \sigma' \in S_3$ are distinct permutations, there exists $n \in \{1, 2, 3\}$, such that $n = \sigma(m) = \sigma'(m')$ for distinct $m, m' \in \{1, 2, 3\}$ (or in other words, because $\sigma \neq \sigma'$, then $\sigma^{-1} \neq \sigma'^{-1}$, hence $m = \sigma^{-1}(n) \neq \sigma'^{-1}(n) = m'$ for some $n \in \{1, 2, 3\}$). WLOG, up to reindexing one can say $m = 1, m' = 2$. Then, notice the two states satisfies the following:

$$\int_{x_1} \int_{x_2} \int_{x_3} (\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)}))^* (\psi_a(x_{\sigma'(1)})\psi_b(x_{\sigma'(2)})\psi_c(x_{\sigma'(3)})) x_3 dx_2 dx_1 = 0 \quad (51)$$

And, the reason is because, x_n in the first (conjugated) term corresponds to $x_{\sigma(m)}$, and the second (non-conjugated) term corresponds to $x_{\sigma'(m')}$, which are inputs of distinct states because $m \neq m'$, causing the two states to be orthogonal. For instance, if $m = 1, m' = 2$, then x_n corresponds to $\psi_a(x_{\sigma(1)})^*$ in the conjugated term, and corresponds to $\psi_b(x_{\sigma'(2)})$ in the non-conjugated term, which the two are orthonormal, hence the integral contains a multiple of $\int_{x_n} \psi_a(x_{\sigma(1)})^* \psi_b(x_{\sigma'(2)}) dx_n$, which is 0.

(On the other hand, if $\sigma = \sigma'$, the above term is clearly 1 due to the three states being normalized in each integral).

Hence, we derived the following:

$$\int_{x_1} \int_{x_2} \int_{x_3} (\psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)}))^* (\psi_a(x_{\sigma'(1)})\psi_b(x_{\sigma'(2)})\psi_c(x_{\sigma'(3)})) x_3 dx_2 dx_1 = \delta_{\sigma\sigma'} \quad (52)$$

As a result, if $\psi_{abc} = A \sum_{\sigma \in S_3} \psi_a(x_{\sigma(1)})\psi_b(x_{\sigma(2)})\psi_c(x_{\sigma(3)})$ (where $A > 0$ is a normalization constant), then it satisfies the following:

$$1 = \int_{x_1} \int_{x_2} \int_{x_3} \psi_{abc}^* \psi_{abc} dx_3 dx_2 dx_1 \quad (53)$$

$$= |A|^2 \sum_{\sigma \in S_3} \sum_{\sigma' \in S_3} \int_{x_1} \int_{x_2} \int_{x_3} (\psi_a(x_{\sigma(1)}) \psi_b(x_{\sigma(2)}) \psi_c(x_{\sigma(3)}))^* (\psi_a(x_{\sigma'(1)}) \psi_b(x_{\sigma'(2)}) \psi_c(x_{\sigma'(3)})) dx_3 dx_2 dx_1 \quad (54)$$

$$= |A|^2 \sum_{\sigma \in S_3} \sum_{\sigma' \in S_3} \delta_{\sigma\sigma'} \quad (55)$$

$$= |A|^2 \sum_{\sigma \in S_3} 1 \quad (56)$$

$$= 6|A|^2 \quad (57)$$

(Note: Recall that S_3 has $3! = 6$ terms).

Hence, for normalization one needs $|A| = \frac{1}{\sqrt{6}}$, or $A = \frac{1}{\sqrt{6}}$. Hence, for identical boson, here is the state:

$$\psi_{abc} = \frac{1}{\sqrt{6}} \sum_{\sigma \in S_3} \psi_a(x_{\sigma(1)}) \psi_b(x_{\sigma(2)}) \psi_c(x_{\sigma(3)}) \quad (58)$$

(c)

□

5

Question 5. *5 Spins in the SHO (similar to Griffiths 5.9 in third edition)*

Consider two non-interacting, identical particles of mass m moving in a one-dimensional simple harmonic oscillator potential of frequency ω . Determine the energies and wavefunctions of the ground state and the first excited state for

- (a) two spin $1/2$ fermions*
- (b) two spin 1 bosons*

Proof.

□

Question 6. *6 Atoms in a box*

Suppose we have a 3D cubic box with sides of length L . The potential is zero inside the box and infinite outside. We place some number N of indistinguishable non-interacting particles of mass m and spin S into the box.

As a reminder, the allowed energies for a single particle in the 1D box are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \equiv n^2 E_1$$

- (a) What is the ground state degeneracy for $N = 3$, and $S = 0$?
- (b) What is the ground state degeneracy for $N = 3$, and $S = 1/2$?
- (c) What is the ground state degeneracy for $N = 2$, and $S = 1$?
- (d) What is the ground state degeneracy for $N = 4$, and $S = 3/2$?

Proof.

□

Question 7. 7 The Ground State of Lithium (Griffiths 5.16, DNE in second edition)

Ignoring electron-electron repulsion, construct the ground state of lithium ($Z = 3$). Start with a spatial wave function, remembering that only two electrons can occupy the hydrogenic ground state; the third goes to $\psi_{2,0,0}$. What is the energy of this state? Now tack on the spin, and antisymmetrize. What's the degeneracy of the ground state?

Hint: Griffiths 5.10 (doesn't exist in second edition) might be very useful.

Proof.

□